

Powering the Future

# GDC Production Data

## QC Reference Guide

---

February 2026

Audience:

Business users performing quality checks on the GDC  
(geoLOGIC/geoSCOUT) monthly well production output tables.

# Table of Contents

---

1. What Are the Output Tables?
2. Where Does the Data Come From?
3. How the Data Flows (Step by Step)
4. How Production Values Are Calculated
5. Calculated (Derived) Products
  - 5.1 Gas Shrinkage Factor
  - 5.2 Field Condensate
6. Unit Conversions
7. Gas Injection Exception Handling
8. Key Filters and Scope
9. Flat Table — Column Groups at a Glance
10. QC Checklist
11. Data Lineage Summary

# 1. What Are the Output Tables?

The production pipeline produces **two final tables**, both at the grain of **one well (UWI) per month**:

| Table                            | Format                      | Description                                                                                                                  |
|----------------------------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------|
| gdc_well_production_monthly_flat | Wide<br>(spreadsheet-like)  | One row per well per month. Every product and unit gets its own column (100+ columns). Includes all wells in the gDC source. |
| gdc_well_production_monthly_long | Long<br>(database-friendly) | One row per well per month per product per unit system. Whitecap wells only. Filters out months with zero volumes.           |

**Key difference:** The flat table includes all wells and all activity types as columns. The long table is scoped to **Whitecap wells only** and structured for analytical queries with product/unit filtering.

# 2. Where Does the Data Come From?

## Primary Source

pden\_vol\_by\_month — This is the raw monthly production volume table from geoLOGIC's PDEN (Production Data Entity) system. It contains well-level volumes reported by product and activity type.

The source data arrives in a **wide format** — one row per well per year, with separate columns for each month (Jan through Dec). The pipeline **unpivots** this into a normalized long format: one row per well, per month, per product, per activity.

## Supporting Inputs

| Input                                                | Purpose                                                                                                                       |
|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| master_well_ticket ( <b>corporate</b> )              | Identifies Whitecap wells (used to filter the long-format table)                                                              |
| ref_unit_conversions ( <b>corporate</b> )            | Stores the conversion factors for metric-to-imperial unit math                                                                |
| gdc_injection_volume_exceptions ( <b>seed file</b> ) | Manually maintained list of specific wells + months where GAS injection data was incorrectly coded in gDC — it's actually CO2 |

### 3. How the Data Flows (Step by Step)

```

gDC pden_vol_by_month (raw, wide format - one row per well/year)
|
v
[1] UNPIVOT to long format (one row per well/month/product/activity)
+ Flag gas injection exceptions from the seed file
|
+-----+
v               v
[2a] PRODUCTION volumes      [2b] INJECTION volumes
(Whitecap wells only)      (Whitecap wells only)
+ Unit conversions          + Unit conversions
+ Gas shrinkage calc        + Gas->CO2 correction
+ Field cond calc           |
+-----+-----+
v
[3] COMBINE production + injection onto single rows
|
+-----+-----+
v               v
[4a] FLAT table              [4b] LONG table
(all wells)                  (Whitecap only)
(all activities              (zero volumes
as columns)                  excluded)

```

#### Step Details

##### Step 1 — Unpivot & Normalize (base\_tmp\_gdc\_\_pden\_well\_vol\_by\_month)

- Converts the 12 monthly columns (jan\_volume...dec\_volume) into individual rows
- Filters to well-level data only (pden\_type = 'PDEN\_WELL')
- Calculates days\_in\_month from the volume date (used later for daily rates)
- Joins the gas injection exception seed to flag specific well+month combinations where GAS is really CO2

##### Step 2a — Production Aggregation (int\_tmp\_gdc\_\_pden\_well\_production\_vol\_by\_product)

- Filters to Whitecap wells via the corporate master well ticket
- Keeps only production-related activities: PRODUCTION, SALES, STORAGE RECOVERY, LOAD RECOVERY
- Keeps key products: HRS, BOE, OIL, GAS, CND, WTR, NGL, ETH, BUT, PRO, PENT, PENT+
- Aggregates volumes by well/month/product
- Calculates both metric and imperial values using stored conversion factors
- Calculates two derived products: **Gas Shrinkage** and **Field Condensate** (see Section 5)

**Step 2b — Injection Aggregation** (int\_tmp\_gdc\_\_pden\_well\_injection\_vol\_by\_product)

- Same Whitecap well filter
- Keeps injection activities: INJECTION, STORAGE INJECTION, DISPOSAL
- Keeps injection products: HRS, INJHRS, INJPRS, WTR, GAS, CO2, CND, STM, SOL, POL
- Applies the gas exception correction: for flagged wells, GAS volumes are reclassified as CO2
- Excludes original CO2 rows for exception wells to avoid double-counting

**Step 3 — Combine** (int\_tmp\_gdc\_\_pden\_well\_combined\_vol\_by\_product)

- Unions production and injection data
- Consolidates onto a single row per well/month/product/unit system
- All volume columns (production, sales, injection, disposal, etc.) appear on the same row

**Step 4a — Flat Table** (gdc\_well\_production\_monthly\_flat)

- Reads the unpivoted base data directly (not the combined intermediate — this covers all wells, not just Whitecap)
- Pivots every product+activity combination into its own named column
- Applies unit conversions inline
- Includes gas exception logic for CO2 injection
- Calculates gas shrinkage factor inline

**Step 4b — Long Table** (gdc\_well\_production\_monthly\_long)

- Reads from the combined intermediate (Whitecap wells only)
- Applies product-specific aggregation rules (see Section 4)
- Calculates daily production and injection rates
- Excludes rows where both production and injection are zero

## 4. How Production Values Are Calculated

Not all products use the same activity types to arrive at a "production" number. The rules are:

| Product                                         | What Counts as "Production"                   | What Counts as "Injection"               |
|-------------------------------------------------|-----------------------------------------------|------------------------------------------|
| Oil                                             | PRODUCTION + STORAGE RECOVERY                 | N/A                                      |
| Gas                                             | PRODUCTION + STORAGE RECOVERY                 | INJECTION + STORAGE INJECTION + DISPOSAL |
| Water                                           | PRODUCTION + STORAGE RECOVERY + LOAD RECOVERY | INJECTION + STORAGE INJECTION + DISPOSAL |
| CO2                                             | PRODUCTION + STORAGE RECOVERY                 | INJECTION + STORAGE INJECTION + DISPOSAL |
| Condensate                                      | PRODUCTION + STORAGE RECOVERY                 | INJECTION + STORAGE INJECTION + DISPOSAL |
| BOE                                             | PRODUCTION + STORAGE RECOVERY                 | N/A                                      |
| NGL, Ethane, Butane, Propane, Pentane, Pentane+ | SALES only (plant products)                   | N/A                                      |
| Hours                                           | PRODUCTION only                               | N/A                                      |
| Polymer, Solvent, Steam                         | N/A                                           | INJECTION + STORAGE INJECTION + DISPOSAL |
| Injection Pressure                              | N/A                                           | MAX value from injection activities      |

**Why the difference?** Water includes load recovery because load fluid recovered at the wellhead is part of total produced water. NGL-family products are only meaningful as sales volumes because they're separated at the plant, not measured at the wellhead.

### Daily Rates

All daily rates are calculated as:  $\text{monthly\_volume} / \text{days\_in\_month}$

## 5. Calculated (Derived) Products

These don't come directly from gDC — they're calculated by the pipeline:

### 5.1 Gas Shrinkage Factor

**What it is:** The proportion of raw gas production that is "lost" (consumed as fuel, vented, flared, or shrunk at the plant) between the wellhead and the sales meter.

$$\text{Gas Shrinkage} = 1 - (\text{Gas Sales Volume} / \text{Gas Production Volume})$$

- A shrinkage of 0.15 means 15% of produced gas doesn't make it to sales
- Returns 0 if either production or sales is zero

### 5.2 Field Condensate

**What it is:** A combined measure of condensate (CND) and pentane-plus (PENT+) volumes, representing total field-level liquid hydrocarbons recovered from gas.

$$\text{Field Condensate} = \text{CND (production + sales)} + \text{PENT+ (production + sales)}$$

## 6. Unit Conversions

All volumes are provided in **both metric and imperial** units. The conversion factors are stored in a corporate reference table and applied consistently:

| Conversion            | From | To    | Notes                                               |
|-----------------------|------|-------|-----------------------------------------------------|
| Oil/Condensate/NGLs   | m3   | bbl   | Standard conditions adjustment included             |
| Water                 | m3   | bbl   | Pure volumetric (different factor than oil)         |
| Gas (hydrocarbon)     | e3m3 | mcf   | Includes Canadian/US standard conditions adjustment |
| Gas (non-HC: CO2, N2) | e3m3 | mcf   | Pure volumetric conversion                          |
| Energy                | GJ   | MMBtu | Thermal energy content                              |
| Pressure              | kPa  | psi   | Injection pressure                                  |

### QC Tip

If metric-to-imperial ratios look off, check the conversion factor columns included in the flat table (`oil_conversion_factor`, `gas_conversion_factor`, etc.). The factors should be consistent across all rows.

## 7. Gas Injection Exception Handling

A small number of wells have **CO2 injection volumes incorrectly coded as GAS** in the gDC source. These are tracked in a manually maintained seed file (`gdc_injection_volume_exceptions.csv`) that lists specific well+month combinations.

### What the pipeline does:

- For flagged wells: GAS injection volumes are **reclassified as CO2 injection**
- The original CO2 rows (if any) for those wells are excluded to prevent double-counting
- In the flat table: flagged wells show 0 for `gas_injection_volume` and the volume appears under `co2_injection_volume` instead

### QC Tip

If you see a well with unusually high CO2 injection or zero gas injection, check whether it appears in the exception list. Currently ~10 wells across 3 date periods are flagged.



## 8. Key Filters and Scope

| Filter                      | Applied Where           | Effect                                            |
|-----------------------------|-------------------------|---------------------------------------------------|
| pden_type = 'PDEN_WELL'     | Base unpivot            | Only well-level data (not facility or pool-level) |
| wcap_ind = 'Y'              | Whitecap well list      | Long table and intermediates are Whitecap-only    |
| volume > 0                  | Intermediate models     | Zeros are excluded from intermediate aggregation  |
| Production OR Injection > 0 | Long table final output | Zero-volume rows excluded from final long table   |
| No filter on wells          | Flat table              | All gDC wells are included                        |

## 9. Flat Table — Column Groups at a Glance

The flat table is organized into these logical sections (100+ columns total):

| Section                   | Example Columns                                                                       | Notes                                |
|---------------------------|---------------------------------------------------------------------------------------|--------------------------------------|
| <b>Identity</b>           | uwi, source, volume_date, days_in_month                                               | One row per well per month           |
| <b>Operating Hours</b>    | hours_on_prod, hours_on_inj                                                           | Production vs injection hours        |
| <b>BOE</b>                | boe_volume, boe_sales_volume,<br>boe_thermal_energy_gj/mmbtu                          | Barrel of oil equivalent             |
| <b>Oil</b>                | oil_volume_m3/bbl, oil_sales_volume_m3/bbl,<br>oil_load_*                             | Production, sales, and load tracking |
| <b>Gas</b>                | gas_volume_e3m3/mcf, gas_sales_*, gas_fuel_*,<br>gas_vented_*, gas_flared_*           | Full gas disposition                 |
| <b>Gas Shrinkage</b>      | gas_shrinkage_factor                                                                  | Calculated ratio                     |
| <b>Water</b>              | water_volume_m3/bbl, water_load_*                                                     | Production and load tracking         |
| <b>Cond &amp; CO2</b>     | cond_volume_*, co2_volume_*, n2_volume_*,<br>field_cond_sales_*                       | Including non-hydrocarbon gases      |
| <b>Plant Products</b>     | ngl_*, butane_*, ethane_*, propane_*, pentane_*,<br>lite_mix_*                        | Sales only (spec, mix, combined)     |
| <b>Injection</b>          | gas_injection_*, water_injection_*,<br>co2_injection_*, polymer_*, solvent_*, steam_* | All injection streams                |
| <b>Pressure</b>           | max_injection_pressure_kpa/psi                                                        | Max monthly injection pressure       |
| <b>Metadata</b>           | activity_type_count, activity_type_list,<br>facility_id_list, last_modified_date      | Audit/traceability info              |
| <b>Conversion Factors</b> | oil_conversion_factor, gas_conversion_factor, etc.                                    | For verifying imperial values        |

## 10. QC Checklist

---

When reviewing the production data, consider these checks:

- **Row counts:** Long table should have fewer rows than flat (Whitecap filter + zero exclusion)
- **Unit consistency:** Pick a well, multiply a metric value by the conversion factor — does it match the imperial column?
- **Gas shrinkage reasonableness:** Values should typically be 0–0.30 (0–30%). Values > 0.5 are unusual and may indicate data issues
- **Gas exception wells:** Verify wells in the exception seed have CO2 injection (not GAS injection) in the output
- **Water production:** Includes load recovery — compare against raw gDC if water seems high for a well
- **NGL products:** Should only appear under sales activity — if you see "production" NGL volumes, something is wrong
- **Zero-month gaps:** The long table excludes zero months, so gaps in a well's timeline may be intentional (no production that month)
- **Duplicate check:** Each well should have exactly one row per month in the flat table, and one row per month per product per unit system in the long table
- **Last modified date:** Compare against gDC to confirm data freshness
- **BOE totals:** BOE is sourced from gDC (not calculated in-house) — discrepancies between oil+gas BOE equivalent and reported BOE may reflect gDC's internal methodology

## 11. Data Lineage Summary

### SOURCES

- └ gDC: pden\_vol\_by\_month — Raw monthly volumes (wide format)
- └ Corporate: master\_well\_ticket — Whitecap well filter
- └ Corporate: ref\_unit\_conversions — Metric↔Imperial factors
- └ Seed: gdc\_injection\_volume\_exceptions — GAS↔CO2 corrections

### PROCESSING

- └ Unpivot wide→long (12 monthly columns → 12 rows)
- └ Filter to well-level data (PDEN\_WELL)
- └ Flag gas exceptions
- └ Split into production vs injection streams
- └ Aggregate by well/month/product
- └ Calculate gas shrinkage + field condensate
- └ Apply unit conversions (metric + imperial)
- └ Combine production + injection

### OUTPUTS

- └ gdc\_well\_production\_monthly\_flat — Wide format, all wells
- └ gdc\_well\_production\_monthly\_long — Long format, Whitecap only